

Note 15.2 Double Integrals over General Regions

Double Integrals over Bounded, Nonrectangular Regions

The Riemann sum are obtained by covering the region with a grid of small rectangular cells whose union contains all points of the region. Partition is formed by taking the rectangles lying completely inside it.

Volume

Theorem 2--Fubini's Theorem(Stronger Form)

For a continuous function $f(x, y)$,

- if $R : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)$ and g_1, g_2 are continuous on $[a, b]$, then

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

- if $R : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)$ and h_1, h_2 are continuous on $[c, d]$, then

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

Note that when reordering the integrals, we need to rewrite the limit of integration by analyzing or sketching rather than simply reorder the limits.

Properties of Double Integrals

Similar to single integrals.

Exercises

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